# Summary

- An event stream represents entities' status updates over time
- The main components of an ESP are Event broker, Event storage, Analytic, and Query Engine
- Apache Kafka is a very popular open-source ESP
- Popular Kafka service providers include Confluent Cloud, IBM Event Stream, and Amazon MSK
- The core components of Kafka are brokers, topics, partitions, replications, producers, and consumers
- The Kafka-console-consumer manages consumers
- Kafka Streams API is a simple client library supporting you with data processing in event streaming pipelines
- A stream processor receives, transforms, and forwards the processed stream
- Kafka Streams API uses a computational graph
- There are two special types of processors in the topology: The source processor and the sink processor

1. Event streams represent entity status updates over time. Events have different formats, which of the most common formats can be either a primitive or complex data type? (multiple answers)

- [ ] Complex format
- [x] Key-value format

  Correct, this format can be a primitive data type or complex data type.

- [ ] Primitive format
- [x] Key-value with a timestamp

  Correct, this key-value format is time-sensitive.

2. How does Kafka increase fault-tolerance and throughput?

- ( ) Indexes
- (•) Topic partitions and replications

  Correct, the topics are partitioned and replicated amongst brokers so if one goes down there are others available.

- ( ) Keys
- ( ) Maps

3. The ad hoc processors that perform stream processing can become complicated. The Kafka Streams API helps solve this complication. How does the Streams API help stream processing? (multiple answers)

- [x] Processes and analyzes data

  Correct, it processes and analyzes data stored in Kafka topic so both input and output are Kafka topics.

- [ ] Processes one record at a time
- [x] Scripts for CLI

  Incorrect, scripts are provided with the command line interface for Kafka.

- [x] Provides client library

  Incorrect, a simple client library is what Streams API uses.

4. Which of the following describes event streaming?

- ( ) Large event volume
- ( ) Observable state updates over time
- (•) Continuous event transportation

  Correct, the continuous event transportation between an event source and event destination is event streaming.

- ( ) External database

5. What was Apache Kafka originally used for?

- ( ) Payments and transactions
- ( ) Data storage
- (•) Track user activities

  Correct, Kafka was originally used to track user activities like keyboard strokes, mouse click, and page views.

- ( ) Auditing

1. The Kafka server side is a cluster with many associated servers. What are the associated servers called? 1

- ( ) Controllers
- ( ) Associates
- ( ) Sub-servers
- (•) Brokers

**Correct**
Correct, Kafka associated servers are called brokers that act as the event broker.

2. Which of the following is Kafka Streams API based on? 1

- ( ) Gantt chart
- (•) Computational graph
- ( ) Java
- ( ) Transformational graph

**Correct**
Correct, the Streams API is based on a computational graph called a stream-processing topology.

**3.** Which of the following do stream processors do?

- ○ Extracts, transforms, and loads
- ○ Processes and forwards
- ○ Extracts, loads, and transforms
- ◉ Receives, transforms, and forwards

> **Correct**
> Correct, stream processors receive, transform, and forward the streams.

**4.** Kafka Streams API is based on a computational graph called a stream processing topology. And in the topology, each node is a stream processor, while edges are the I/O streams. In this topology we find two special types of processors: What are they called?

- ○ Aggregation and stream processor
- ◉ Source and sink processor
- ○ Mapping and transformation processor
- ○ Stream and topic processor

> **Correct**
> Correct, there are two special types of processors in the topology: The source processor and the sink processor.

5. Which of the following Kafka main features provides consumption without a deadline?

- ○ Reliability
- ◉ Permanent persistency
- ○ Distribution system
- ○ Open source

**Correct**
Correct, Kafka stores events permanently so consumers can access streaming events at any time.

6. Once events are published and properly stored in topic partitions, you can create _________ to read them.

- ○ Producers
- ○ Brokers
- ◉ Consumers
- ○ Partitions

**Correct**
Correct, once events are published and properly stored in topic partitions, you can create consumers to read them.

7. The core component of any ESP is the event broker. Which event broker sub-component performs encryption on data?

- ○ Ingester
- ○ Storage
- ◉ Processor
- ○ Consumption

**Correct**
Correct, the processor performs operations on data like serializing, compressing, and encryption.

8. ESPs are a middle layer between multiple event sources and destinations. ESPs may have different architectures and components but also some common components. Which of the following common components receives and consumes events?

- ◉ Event broker
- ○ Event storage
- ○ Analytic engine
- ○ Query engine

**Correct**
Correct, this is the core component of an ESP that receives and consumes events.

9. Which of the following Kafka core components publish events into topics?

- ○ Consumers
- ○ Partitions
- ◉ Producers
- ○ Brokers

**Correct**
Correct, these are client applications that publish events into topics.

10. Which of the Kafka CLI script files manages topics?

- ○ Kafka-console-producer
- ◉ Kafka-topics
- ○ Kafka-console-consumer
- ○ Kafka-console

**Correct**
Correct, this CLI manages topics.